Alyssa Xiaole Liu

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Department of Supply Chain Management
Rutgers Business School-Newark and New Brunswick, Rutgers University

Academic Appointments

2026 – Present **Rutgers University, Rutgers Business School**, Newark and New Brunswick, NJ
Assistant Professor

Education

2026 **New York University Stern School of Business**, New York, NY
Ph.D. in Operations Management
Committee: Mor Armony (advisor), Jing Dong (advisor), Rouba Ibrahim (advisor), Jackie Baek, and Michael L. Pinedo

2020 **Columbia University**, New York, NY
M.Sc. in Operations Research

2018 **University of Michigan-Dearborn**, MI
M.S.E. in Industrial and System Engineering

2016 **Xi'an Jiaotong University**, Xi'an, Shaanxi, China
B.Mgt. in Management Science and Engineering
The Special Class for the Gifted Young of China (XJTU)

Research Interests

Healthcare operations; AI-enabled and data-driven decision making and policy learning; causal inference and machine learning in operations; stochastic modeling; technology adoption and healthcare delivery innovation; efficiency and fairness; empirical and applied research in collaboration with healthcare providers.

Papers

2025 **Measuring the Heterogeneous Effect of Emergency Department Boarding on Inpatient Length of Stay: Toward Efficient and Equitable Inpatient Bed Assignment** [Link]
with Jing Dong, Yosef Berlyand, Martin S. Copenhaver.
Major Revision Resubmitted, *Management Science*.

2026 **Proactive Transfer Admission Control for Emergency Departments** [Link]
with Ruicheng Ao, Jing Dong, Martin S. Copenhaver.
Major Revision Resubmitted, *Manufacturing & Service Operations Management*.

2024 **Telemedicine versus In-Person Outpatient Care: Equilibrium, Capacity, and Quality Differentiation** [Link]
with Mor Armony.
Major Revision, *Manufacturing & Service Operations Management*.
Accepted to the M&SOM Service Management SIG 2025.

2025 **Fair and Efficient Outpatient Scheduling with Stratified No-Show Prediction**
with Jing Dong, Rouba Ibrahim.
In Preparation for Submission, *Available upon Request*.

2019 **Private and Public Incentives for Hospitals to Improve the Quality and Reduce the Cost of Care** [Link]
with Kevin Schulman, David Scheinker.
Health Management Policy and Innovation, Volume 4, Issue 3.

Research In Progress

From Overflow to Oversight: The Strategic Role of Hallway Beds in Hospital Patient Flow
Xiaole (Alyssa) Liu, Jing Dong, Martin S. Copenhaver.

Awards and Honors

2026 **Herman E. Krooss Award**, NYU Stern
2026 **SCAI Annual Conference PhD Dissertation Awards (3rd Place)**, Rutgers
2024 **W. Edwards Deming Fellowship**, NYU Stern
2024 **Ph.D. Alumni Fellowship**, NYU Stern
2024 **Teaching Commendation**, NYU Stern Doctoral Program
2020 **Doctoral Fellowship** for 2020-2025, NYU Stern
2020 **Operations Research Graduate Fellowship**, Columbia University
2018 **Honor Scholar**, UM-Dearborn
2016 **Graduate Student Instructor Fellowship** for 2016-2018, UM-Dearborn
2014 **Suzhou Industrial Park Scholarship**, XJTU
2013 **National Motivational Scholarship**, XJTU, Ministry of Education China

Teaching Experience

Instructor

2027 33.799.310: Demand Planning (Rutgers Business School undergraduate core)

2024	OPMG-UB.1: Operations Management (NYU Stern undergraduate core) Teaching evaluation: 4.8/5.0 *with Teaching Commendation*.
	Teaching Assistant
2023, 25, 26	COR1-GB.2114: Operations (NYU Stern M.Sc.)
2022	SHBI-GB.7301: Stochastic Modeling & Simulation (NYU Stern M.Sc.)
2022	OPMG-UB.1: Operations Management (NYU Stern undergraduate core)
2020	IEOR E4729: Model-Based Trading: Theory and Practice (Columbia M.Sc.)
2019	IEOR E4150: Introduction to Probability and Statistics (Columbia M.Sc.)
2016 – 2018	IMSE 561: Total Quality Management and Six Sigma (UM-Dearborn M.S.E.)
2017	IMSE 4585: Simulation in Systems Design (UM-Dearborn undergraduate)
2016	IMSE 511: Design and Analysis of Experiment (UM-Dearborn M.S.E.)

Invited Talks

2026	Rotman Young Scholar Seminar Series, OM Department, University of Toronto
2026	SCAI Annual Conference PhD Dissertation Competition, Rutgers, Newark, NJ
2026	Rutgers University, Rutgers Business School, Piscataway, NJ
2026	W. P. Carey School of Business at Arizona State University, Tempe, AZ
2026	Cornell SC Johnson College of Business, Cornell University, Ithaca, NY
2026	C. T. Bauer College of Business, University of Houston, Houston, TX
2026	The University of Chicago Booth School of Business, Chicago, IL
2025	Michigan State University Broad College of Business, East Lansing, MI
2025	ETH Zurich, D-MTEC, Zurich, Switzerland

Selected Presentations

	Measuring the Heterogeneous Effect of Emergency Department Boarding on Inpatient Length of Stay: Toward Efficient and Equitable Inpatient Bed Assignment
2026	INFORMS Anuual Meeting, San Francisco, CA
2026	IMSI Workshop *Advances in Quantitative Medical Care* Lightning Talk, Chicago, IL
2025	INFORMS Annual Meeting, Atlanta, GA
2024	INFORMS Annual Meeting, Seattle, WA
2024	M&SOM Conference, Minneapolis, MN
	Telemedicine versus In-Person Outpatient Care: Equilibrium, Capacity, and Quality Differentiation
2025	M&SOM Service Management SIG, London, UK
2023	INFORMS Annual Meeting, Phoenix, AZ
2023	M&SOM Conference, Montreal, Canada

2023	POMS Conference, Orlando, FL
2022	INFORMS Annual Meeting, Indianapolis, IN
2021	INFORMS Annual Meeting, virtual

Professional Services

Ad-hoc Reviewer	Management Science, Manufacturing & Service Operations Management, Omega, M&SOM Conference, DSI Emerging Economies Research Award
2026 – Present	Co-chair: Supply Chain Management Department Seminar Series, Rutgers
2024	Co-organizer: NYC Operations Day Joint PhD Colloquium
2022	Organizer: MOILS Seminar Series, NYU Stern

Research Experience and Industry Collaboration

2023 – Present **Massachusetts General Hospital**, Boston, MA
Researcher, Healthcare Systems Engineering.

2022 **Waitwhile**, San Francisco, CA
Research Consultant, Queue Management Software for the Modern Enterprise.

2019 **Stanford University**, Stanford, CA
Research Assistant, Systems Utilization Research For (SURF) Stanford Medicine.

Industry Work Experience

2018 **Stellantis (Fiat Chrysler Automobile)**, Auburn Hills, MI
Full-time Simulation Engineer.

2015 **JD.com, Inc.**, Xi'an, Shaanxi, China
Summer Intern, Inventory Management.

References

Mor Armony
Vice Dean for Faculty and Research
Harvey Golub Professor of Business Leadership
Professor of Technology, Operations & Statistics
Leonard N. Stern School of Business
New York University
Email: marmony@stern.nyu.edu

Jing Dong
DeRosa Family Associate Professor of Business
Decision, Risk, and Operations Division
Graduate School of Business
Columbia University
Email: jing.dong@gsb.columbia.edu

Rouba Ibrahim
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School of Management
University College London (UCL)
Email: rouba.ibrahim@ucl.ac.uk

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Center for Research and Evaluation
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Massachusetts General Hospital
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